A Decomposition Approach for Evaluating Security of Masking

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Abstract. Masking is a commonly used countermeasure against side-channel attacks, encoding secrets into multiple shares such that each share leaks only partial information. A longstanding question is under what noise conditions masking guarantees security, and how this security scales with the number of shares. While sufficient conditions have been known for binary fields and in high-noise regimes, the borderline and low-noise cases have remained poorly understood.
In this work, we close this gap through a decomposition-based analysis. Our approach reduces leakage in extended fields to binary projections, enabling tight bounds on the adversary's success rate and yielding an optimal reduction from the noisy leakage model to the random probing model—even in regimes where classical reductions fail. As a central theoretical result, we prove a conjecture of Dziembowski et al. (TCC 2016), showing that for any additive group $\mathbb{G}$ with largest proper subgroup $\mathbb{H}$, masking strictly improves security whenever the leakage is δ-noisy with $\delta < 1 - \frac{|\mathbb{H}|}{|\mathbb{G}|}$.
We additionally demonstrate the practical relevance of our framework for leakage certification and for determining the required masking order under realistic low-noise conditions. Our results unify and sharpen the understanding of noise requirements for masking, advancing both the theoretical foundations and the practical evaluation of side-channel countermeasures.

1 Introduction

Masking to Mitigate Side-Channel Threats. Side-channel information refers to unintended physical leakages that adversaries can exploit from cryptographic implementations. Prominent examples include variations in power consumption, electromagnetic emanations, or timing behavior. To reason about such leakages formally, the *noisy leakage model* was introduced by Prouff and Rivain [28] and subsequently explored in a series of works [10, 11, 13, 14, 27]. In this model, for each intermediate value $X \in \mathbb{F}_q$, the adversary observes a randomized function $\mathsf{L}(X)$, such as a noisy Hamming weight.

A widely deployed countermeasure against such leakages is *masking*. In masking, a sensitive variable X is split into random *shares* $X_1, \ldots, X_n \in \mathbb{F}_q$ such that

$X = X_1 + \cdots + X_n$. The implementation then manipulates only the shares, so that the adversary's leakage consists of observations $\mathsf{L}(X_i)$ on each share, rather than on the secret itself. In a standalone encoding, the shares are the only intermediates, while in a full protected circuit multiple secrets interact, leading to many intermediate values. The effectiveness of masking is usually quantified by how a *security metric*—for example, the adversary's success probability—degrades as the number of shares n increases. Since the seminal work of Chari et al. [8], this methodology has shaped both theoretical studies and practical evaluations of masking-based countermeasures.

Open Challenges. If the leakage function $\mathsf{L}(X)$ reveals the full value of X, then masking offers no security benefit. Thus, some intrinsic amount of noise is required for masking to be effective. The central open question is: *what is the minimal noise level that suffices, and how does the resulting security scale with the number of shares n?* Existing analyses have identified sufficient conditions, but a complete characterization remains elusive—especially in the delicate *borderline regimes* where noise is low. Furthermore, circuit-level security proofs often rely on reductions to other leakage models, such as the random probing model, but these reductions become loose or even meaningless in low-noise conditions. Our work addresses these unresolved challenges.

Practical Relevance. The questions are not only of theoretical interest but also of practical relevance. Low-noise (high-SNR) leakages arise naturally in several settings: *(i)* embedded processors such as the ARM Cortex-M0 have been shown to produce very precise power-consumption patterns with limited noise [7]; *(ii)* static power analysis captures stable signals over long time windows, yielding much cleaner measurements than traditional dynamic power analysis [26]; and *(iii)* horizontal or averaging attacks combine repeated leakage samples of the same intermediate to reduce measurement noise and enhance adversarial advantage [3]. In all these cases, the assumptions underlying earlier noise thresholds become too weak, and stronger and tighter conditions are required.

Security Metrics. Several metrics have been proposed to evaluate the informativeness of leakage. The *success rate (SR)* [30] measures the probability that the adversary correctly identifies the secret from the leakage. The *statistical distance (SD)*, denoted with δ, quantifies the deviation between the prior distribution of X and the posterior distribution given $\mathsf{L}(X)$. The *mutual information (MI)* captures the amount of information about X revealed by $\mathsf{L}(X)$. These measures are theoretically connected—for example, Pinsker's inequality links MI and SD [16], and SR is bounded in terms of SD (see Lemma 1). However, the bounds are typically loose, making it difficult to translate conditions expressed in one metric into tight guarantees for another. In this work, we focus on SR, which is not only intuitive but also directly captures the number of traces an adversary needs for a successful attack. This is particularly relevant in divide-and-conquer strategies, where every chunk of the secret must be guessed correctly in one attempt.

1.1 Related Work and Limitations

Single Encodings. The foundational reduction of Duc et al. [10, 11] showed that $q\delta < 1$ suffices for masking security. Dziembowski et al. [14] sharpened this result by proving that, in binary extension fields, the bound $\delta < 1/2$ is sufficient. Nonetheless, the case where $\delta \geq 1/2$, remained unresolved.

Later works considered MI as a security metric: Ito et al. [21] proved that $\mathsf{MI}(X; \mathsf{L}(X)) < 0.72$ suffices, while Bégouinot et al. [4] allowed ≤ 1 for some shares, relaxing the uniformity assumption. Despite these advances, none of the existing criteria provided a tight and general characterization valid across different leakage models and algebraic structures.

Protected Circuits. For full protected circuits, Prouff and Rivain [28] initiated the study of masking under noisy leakage, later refined by Masure and Standaert [25]. However, proofs of circuit-level security often rely on auxiliary assumptions, such as the presence of leak-free refresh gadgets. A key breakthrough came from Duc et al. [10], who proposed a reduction from the noisy leakage model to the *random probing model* (RPM). This reduction enabled transferring security guarantees but bounded the probing probability by $q\delta$, which grows quickly with q and becomes vacuous when δ is close to 1. More recently, Brian et al. [6], building on work of Dziembowski et al. [13], introduced the *average probing model*, which eliminates the dependence on field size. Yet their approach currently applies only to first order masking, leaving higher-order masking out of reach.

1.2 Our Contributions

This paper develops a *decomposition-based framework* for analyzing masking security, enables tight reductions, and closes several longstanding gaps. Our contributions are:

- **Tight characterization of noise thresholds.** We prove a conjecture of Dziembowski et al. (TCC 2016): for an additive group $\mathbb{G}$ with largest proper subgroup $\mathbb{H}$, any δ-noisy leakage with $\delta < 1 - \frac{|\mathbb{H}|}{|\mathbb{G}|}$ ensures that masking strictly improves security. This provides, for the first time, necessary and sufficient conditions for the effectiveness of masking across both binary and non-binary settings.
- **Decomposition technique for binary extensions.** We introduce a bitwise projection method that reduces the analysis of leakage in $\mathbb{F}_{2^u}$ to binary subproblems. This allows us to derive tight lower and upper bounds on the success rate of masked encodings and gadgets. Our results show that this decomposition yields an optimal reduction from noisy leakage to the RPM, even in low-noise regimes where classical bounds collapse.
- **Generality and practical relevance.** We extend our framework beyond binary settings, analyzing masking over prime fields and general additive groups. On the practical side, we demonstrate how our refined metrics improve leakage certification, and how they can be used to determine the required masking order to meet specific security targets. Our experiments on

real devices confirm that the decomposition-based criteria provide accurate and robust guidance for evaluating masking.

Outline. In Section 2, we introduce the security metrics and their relationships for single variables. In Section 3, we develop our decomposition-based masking analysis in binary extension fields, and in Section 4, we extend the approach to prime groups. In Section 5, we generalize the results to arbitrary additive groups, proving the conjecture of Dziembowski et al. Finally, in Section 6, we apply the decomposition framework to linear masked circuits and illustrate its practical implications.

2 Side-Channel Security of a Single Variable

This section formalizes the noisy leakage model and introduces the adversary's *advantage*, a normalized version of the success rate. We relate this metric to the statistical distance δ in Lemma 1. Subsection 2.2 presents the reduction to the random probing model, and Lemma 2 proves its tightness in binary fields. Finally, Lemma 3 derives a bound on the advantage using this reduction.

2.1 Preliminaries

Let X be uniformly distributed over $\mathbb{F}_q$, representing an intermediate value in a cryptographic implementation. Side-channel leakage is modeled by a probabilistic function $\mathsf{L}(X) \in \mathbb{R}^m$, and the adversary's goal is to guess X from $\mathsf{L}(X)$. The optimal strategy is *maximum a posteriori* (MAP) estimation [20]:

$$\hat{X} \xleftarrow{\$} \left\{ \underset{\alpha \in \mathbb{F}_q}{\operatorname{argmax}} \Pr(X = \alpha \mid l) \right\}, \quad \text{where } l \leftarrow \mathsf{L}(X).$$

The MAP success probability varies with the leakage value. For instance, if $\mathsf{L}(X) = \mathsf{HW}(X)$, then observing $l = 0$ reveals X completely.

Define the average success probability:

$$P_c \triangleq \mathop{\mathbb{E}}_{l}[\Pr(\hat{X} = X \mid l)] = \sum_{l} \Pr(\mathsf{L}(X) = l) \cdot \Pr(\hat{X} = X \mid l),$$

and the *advantage* over random guessing as $\mathsf{Adv}_X \triangleq P_c - \frac{1}{q}$.

Statistical Distance and δ-Noisy Leakage. The *statistical distance* between X and its posterior $X \mid \mathsf{L}(X)$ measures leakage informativeness [28]. It is defined as:

$$\mathsf{SD}(X\,;\, X \mid \mathsf{L}(X)) \triangleq \sum_{l} \Pr(\mathsf{L}(X) = l) \cdot \mathsf{TV}(X\,;\, X \mid l),$$

where the *total variation distance* is:

$$\mathsf{TV}(X\,;\, X \mid l) \triangleq \frac{1}{2} \sum_{\alpha \in \mathbb{F}_q} \left| \Pr(X = \alpha \mid l) - \frac{1}{q} \right| = \sum_{\Pr(X=\alpha \mid l) > \frac{1}{q}} \left(\Pr(X = \alpha \mid l) - \frac{1}{q} \right).$$

We say $\mathsf{L}(X)$ is δ-noisy if $\mathsf{SD}(X\,;\, X \mid \mathsf{L}(X)) = \delta$. By definition, $\delta \in [0, 1 - \frac{1}{q}]$.

Relation Between δ and Adv_X. A lower δ corresponds to higher noise and thus lower advantage. The following lemma formalizes this relationship.

Lemma 1. *Let $X \in \mathbb{F}_q$ and let $\mathsf{L}(X)$ be δ-noisy. Then:*

$$\frac{\delta}{q-1} \leq \mathsf{Adv}_X \leq \delta.$$

In the binary case ($q = 2$), we have $\mathsf{Adv}_X = \delta$.

Proof. For each l, we have $\max_\alpha \Pr(X = \alpha \mid l) - \frac{1}{q} \leq \mathsf{TV}(X\,;\,X \mid l)$. Hence, averaging over l proves the upper bound. The lower bound follows by noting that at most $q-1$ values can exceed $\frac{1}{q}$. $\square$

2.2 Leakage Simulation

The *erasure channel* [9,19] is a probabilistic mapping $\phi^\epsilon \colon \mathbb{F}_q \to \{\perp, \mathbb{F}_q\}$ defined as:

$$\phi^\epsilon(X) = \begin{cases} X & \text{with probability } \epsilon, \\ \perp & \text{otherwise.} \end{cases}$$

Duc et al. [10] showed that for sufficiently noisy leakage $\mathsf{L}(X)$, one can construct a function L' such that for any X, the leakages $\mathsf{L}'(\phi^\epsilon(X))$ and $\mathsf{L}(X)$ are statistically indistinguishable:

$$\forall \alpha \in \mathbb{F}_q, \quad \mathsf{TV}\left(\mathsf{L}'(\phi^\epsilon(\alpha))\,;\,\mathsf{L}(\alpha)\right) = 0.$$

This holds if $\epsilon \geq \epsilon_{\min}$, where

$$\epsilon_{\min} \triangleq 1 - \sum_l \min_{\alpha \in \mathbb{F}_q} \Pr(l \mid X = \alpha) \ \leq_{(\mathtt{I})}\ q\delta. \tag{1}$$

This value, known as the *Doeblin coefficient* [5], lies in $[0,1]$: $\epsilon_{\min} = 0$ means $\mathsf{L}(X)$ is independent of X, while $\epsilon_{\min} = 1$ implies that the noise of leakage is too low, and the technique is not applicable.

The right-hand side of inequality $(\mathtt{I})$ was proved in [10]. We now show that in the binary case, equality holds.

Lemma 2. *Let $(X, \mathsf{L}(X))$ be a joint distribution with $\mathsf{SD}(X\,;\,X \mid \mathsf{L}(X)) = \delta$ and X uniform over $\mathbb{F}_2$. Then:*

$$\epsilon_{\min} = 2\delta.$$

Proof.

$$\begin{aligned}
\epsilon_{\min} &= 1 - \sum_l \min_{\alpha \in \{0,1\}} \Pr(l \mid X = \alpha) \\
&= \sum_l \Pr(\mathsf{L}(X) = l)\left[1 - 2\min_\alpha \Pr(X = \alpha \mid l)\right] \\
&= \sum_l \Pr(l)\left[\max_\alpha \Pr(X = \alpha \mid l) - \min_\alpha \Pr(X = \alpha \mid l)\right] \\
&= \sum_l \Pr(l)\left[\left|\Pr(X = 1 \mid l) - \tfrac{1}{2}\right| + \left|\Pr(X = 0 \mid l) - \tfrac{1}{2}\right|\right] = 2\delta. \quad \square
\end{aligned}$$

Since $\delta \le \frac{1}{2}$ for binary X, we have $\epsilon_{\min} < 1$ unless $\mathsf{L}(X)$ fully reveals X. Thus, in the binary case, the residual uncertainty about X is precisely reflected in $\epsilon_{\min}$.

Example 1. Let $X \in \mathbb{F}_{2^u}$ and consider $\mathsf{L}(X) = x_0 \oplus e$, where x_0 is the least significant bit of X and $\Pr(e = 1) = \mathsf{e} \le \frac{1}{2}$. Then $\mathsf{L}(X)$ leaks noisy information about the LSB only. The posterior of X satisfies:

$$\Pr(X = \alpha \mid l) = \begin{cases} \frac{1-\mathsf{e}}{2^{u-1}} & \text{if } \alpha \in \{0,1\}^{u-1}||l, \\ \frac{\mathsf{e}}{2^{u-1}} & \text{otherwise.} \end{cases}$$

Hence:

$$\mathsf{Adv}_X = \frac{1-2\mathsf{e}}{2^u}, \quad \mathsf{SD}(X\,;\,X \mid \mathsf{L}(X)) = \frac{1}{2} - \mathsf{e}, \quad \epsilon_{\min} = 1 - 2\mathsf{e}. \quad \square$$

A Security Reduction. Since $(X, \mathsf{L}(X))$ and $(X, \mathsf{L}'(\phi^\epsilon(X)))$ are identically distributed, the adversary's advantage remains unchanged:

$$\mathsf{Adv}_X[\mathsf{L}(X)] = \mathsf{Adv}_X[\mathsf{L}'(\phi^\epsilon(X))].$$

We use $\mathsf{Adv}_X[\cdot]$ to specify the leakage source explicitly.

The informativeness of the random variables (RVs) $\phi^\epsilon(X)$, $\mathsf{L}'(\phi^{\epsilon_{\min}}(X))$, and $\mathsf{L}(X)$ about X can be expressed with the following chain:

$$X \to \phi^\epsilon(X) \to \phi^{\epsilon_{\min}}(X) \to \mathsf{L}'(\phi^{\epsilon_{\min}}(X)) \to \mathsf{L}(X),$$

with $\epsilon \ge \epsilon_{\min}$. As we move right along the chain, we obtain progressively degraded views of X, so any leakage evaluation metric (e.g., success probability) must decrease:

$$\mathsf{Adv}_X[\mathsf{L}(X)] \le \mathsf{Adv}_X[\phi^\epsilon(X)], \qquad \mathsf{SD}(X\,;\,X \mid \mathsf{L}(X)) \le \mathsf{SD}(X\,;\,X \mid \phi^\epsilon(X)). \quad (2)$$

Thus, to prove security under leakage $\mathsf{L}(X)$, it suffices to prove it under $\phi^{\epsilon_{\min}}(X)$. This reduction from δ-noisy to ϵ-random probing leakage was introduced in [10]. The following lemma is a concrete application.

Lemma 3 ([5], Proposition 1). *Let $X \in \mathbb{F}_q$, and let $\mathsf{L}(X)$ be a leakage with Doeblin coefficient $\epsilon_{\min}$. Then:*

$$\mathsf{Adv}_X \le \frac{q-1}{q} \cdot \epsilon_{\min}.$$

$\epsilon_{\min}$ Is Not Always Tight. While the reduction form δ-noisy to the ϵ-random probing is powerful, it is not always tight. For instance, when $\mathsf{L}(X) = \mathsf{HW}(X)$, equation (1) gives $\epsilon_{\min} = 1$, even though the Hamming weight does not fully determine X. In this case, the reduction becomes ineffective.

One might attribute this to $\mathsf{HW}(X)$ being highly informative. However, consider a simpler leakage function $\mathsf{ZV}(X)$, inspired by the zero-value model [23], defined as:

$$\mathsf{ZV}(X) = \begin{cases} \nu_a & \text{if } X = 0, \\ \nu_b \neq \nu_a & \text{otherwise.} \end{cases} \tag{3}$$

This function merely indicates whether $X = 0$, revealing only one bit of information. Yet, it still results in $\epsilon_{\min} = 1$.

Our refined reduction approach, introduced in the next section, addresses this limitation by enabling a more fine-grained analysis for such cases.

3 Security of Mask Encoding

3.1 Mask Encoding

A standard method for protecting a sensitive variable X is *masking*, where X is split into a random tuple of n shares $\boldsymbol{X} = (X_1, \ldots, X_n)$ such that

$$X = \sum_{i=1}^{n} X_i.$$

The adversary then observes the leakage vector

$$\boldsymbol{L}(\boldsymbol{X}) = \left[\mathsf{L}_1(X_1),\, \ldots,\, \mathsf{L}_n(X_n)\right].$$

We assume for simplicity that all leakage functions are identical, i.e., $\mathsf{L}_i = \mathsf{L}$, and independent. A key question is how the adversary's advantage $\mathsf{Adv}_X[\boldsymbol{l} \leftarrow \boldsymbol{L}(\boldsymbol{X})]$ depends on the number of shares n and the structure of the field $\mathbb{F}_q$.

Intractability of Exact Metrics. The space of possible leakage vectors $\boldsymbol{L}(\boldsymbol{X})$ grows exponentially with the sharing order n, rendering the exact evaluation of security metrics computationally infeasible. A common workaround—adopted also in this work—is to approximate these metrics based on the distribution of $\mathsf{L}(X)$, albeit at the cost of potential inaccuracies.

A Loose Bound. Duc et al. [11] applied the δ-noisy to ϵ-random probing reduction, in which each share is revealed with probability $\epsilon \leq q\delta$. In the random probing model, the adversary learns each share independently with probability ϵ, so the probability of learning all n shares is at most $(q\delta)^n$. The corresponding leakage vector in this model is:

$$\boldsymbol{\phi}^{\boldsymbol{\epsilon}}(\boldsymbol{X}) = \left[\phi^{\epsilon}(X_1), \ldots, \phi^{\epsilon}(X_n)\right].$$

Generalizing from (2), the statistical distance satisfies:

$$\Delta = \mathsf{SD}(X\,;\, X \mid \boldsymbol{L}(\boldsymbol{X})) \leq \mathsf{SD}(X\,;\, X \mid \boldsymbol{\phi}^{\boldsymbol{\epsilon}}(\boldsymbol{X})).$$

If at least one share is not revealed, the posterior distribution of X is uniform, yielding zero statistical distance. Only when all shares are leaked does the distance reach its maximum value of $1 - \frac{1}{q}$. Therefore, we obtain the bound:

$$\Delta \leq \left(1 - \tfrac{1}{q}\right) q^n \delta^n. \tag{4}$$

A Tighter Bound. The bound in (4) increases rapidly with q, yet empirical results in [11] showed no such dependency. This led to the conjecture that the q-factor may be a proof artifact. A tighter bound, removing this dependency, was later proven by Masure et al. [24]:

Lemma 4 ([24], Proposition 4). *Let $X = (X_1, \ldots, X_n)$ be a masking of $X \in \mathbb{F}_q$, and suppose $\mathsf{SD}(X\,;\, X \mid \mathsf{L}(X)) = \delta$. Then*

$$\Delta = \mathsf{SD}(X\,;\, X \mid \boldsymbol{L(X)}) \leq 2^{n-1}\delta^n.$$

This result confirms an observation of Dziembowski et al. [14]: when $\delta < \frac{1}{2}$, the posterior distribution of X becomes increasingly uniform as n increases. In this setting, the adversary's best guess converges to $\frac{1}{q}$, and applying Lemma 1 yields:

$$\mathsf{Adv}_X[\boldsymbol{l} \leftarrow \boldsymbol{L(X)}] \leq 2^{n-1}\delta^n.$$

Case of $q = 2$. We prove that for binary fields the given bound is tight.

Lemma 5. *In the setting of Lemma 4, if the underlying field is $\mathbb{F}_2$, then*

$$\mathsf{Adv}_X[\boldsymbol{l} \leftarrow \boldsymbol{L(X)}] = 2^{n-1}\delta^n.$$

Proof. We extend a technique by Wyner [31], originally developed for *wiretap channels*.

To recover $X = X_1 \oplus \cdots \oplus X_n$ from the leakage vector $\boldsymbol{L(X)}$, the adversary estimates each X_i individually. Let $\hat{X}_i$ denote the estimate of X_i based on $\mathsf{L}(X_i)$, and let $\mathsf{e}_i = \Pr(\hat{X}_i \neq X_i)$ be the average error probability, taken over both the uniform choice of $X_i \in \{0, 1\}$ and the leakage randomness:

$$\mathsf{e}_i = \mathbb{E}_{X_i,\, l \leftarrow \mathsf{L}(X_i)}\left[\Pr(\hat{X}_i \neq X_i \mid l)\right].$$

Assume $\mathsf{e}_i \leq \frac{1}{2}$ (the case $\mathsf{e}_i > \frac{1}{2}$ is similar). The corresponding advantage is then

$$\mathsf{Adv}_{X_i} = (1 - \mathsf{e}_i) - \tfrac{1}{2} = \tfrac{1}{2} - \mathsf{e}_i.$$

By Lemma 1, we have $\mathsf{Adv}_{X_i} = \delta$, implying $\mathsf{e}_i = \frac{1}{2} - \delta$. Let $\mathsf{e} = \frac{1}{2} - \delta$ for simplicity.

The adversary computes $\hat{X} = \hat{X}_1 \oplus \cdots \oplus \hat{X}_n$. This estimate equals the true value X if an even number of errors occur. Thus, the success probability is:

$$\begin{aligned}
\Pr(\hat{X} = X) &= \sum_{j=0}^{\lfloor n/2 \rfloor} \binom{n}{2j} \mathsf{e}^{2j}(1-\mathsf{e})^{n-2j} \\
&= \frac{1}{2}\left[\sum_{i=0}^{n}\binom{n}{i}\mathsf{e}^i(1-\mathsf{e})^{n-i} + \sum_{i=0}^{n}\binom{n}{i}(-\mathsf{e})^i(1-\mathsf{e})^{n-i}\right] \\
&=_{(\mathsf{I})} \frac{1}{2}\left[(\mathsf{e} + 1 - \mathsf{e})^n + (-\mathsf{e} + 1 - \mathsf{e})^n\right] \\
&= \frac{1}{2}\left[1^n + (1 - 2\mathsf{e})^n\right] = \frac{1}{2} + 2^{n-1}\delta^n,
\end{aligned}$$

where step (I) uses the binomial expansion:

$$(\pm\mathsf{e} + (1-\mathsf{e}))^n = \sum_{i=0}^{n} \binom{n}{i} (\pm\mathsf{e})^i (1-\mathsf{e})^{n-i}.$$

Subtracting the baseline guessing probability $\frac{1}{2}$, we get:

$$\mathsf{Adv}_X[\boldsymbol{l} \leftarrow \boldsymbol{L}(\boldsymbol{X})] = \Pr(\hat{X} = X) - \tfrac{1}{2} = 2^{n-1}\delta^n. \qquad \square$$

The binary case forms the basis for our reasoning over extended fields. Before proceeding, we highlight an important observation.

Optimality of the Reduction at $q = 2$. Lemma 2 establishes that the δ-noisy to ϵ-random reduction is tight for $q = 2$, with $\epsilon_{\min} = 2\delta$. Substituting this into (4), we obtain:

$$\Delta \le \mathsf{SD}(X\,;\, X \mid \boldsymbol{\phi}^{\boldsymbol{\epsilon_{\min}}}(\boldsymbol{X})) = \left(1 - \tfrac{1}{2}\right) \cdot 2^n\delta^n = 2^{n-1}\delta^n.$$

On the other hand, Lemma 5 gives $\Delta = 2^{n-1}\delta^n$, implying that

$$\Delta = \mathsf{SD}(X\,;\, X \mid \boldsymbol{\phi}^{\boldsymbol{\epsilon_{\min}}}(\boldsymbol{X})).$$

This equality shows that the reduction is tight not only for a single variable, but also in the context of mask encoding.[3]

Furthermore, when the leakage functions differ across shares—each with corresponding parameter δ_i—the bound generalizes to:

$$\Delta = 2^{n-1} \prod_{i=1}^{n} \delta_i.$$

For $q > 2$, the Bound in Lemma 4 is Loose We now present a concrete example for $q = 4$, illustrating that the bound in Lemma 4 is not tight in general.

Example 2. Let $X \in \mathbb{F}_{2^2}$, and define the leakage function as

$$\mathsf{L}(X) = (x_1 \oplus e_1) \,\|\, (x_0 \oplus e_0),$$

where x_0, x_1 are the bits of X, and e_0, e_1 are independent Bernoulli variables with $\Pr(e_i = 1) = \mathsf{e} < \frac{1}{2}$. A direct computation yields:

$$\delta = \mathsf{SD}(X\,;\, X \mid \mathsf{L}(X)) = \left(\tfrac{1}{2} - \mathsf{e}\right)\left(\tfrac{3}{2} - \mathsf{e}\right). \tag{5}$$

This leakage is effectively the concatenation of two independent binary leakages. From previous results, we know that under such a leakage model, a masked bit is recovered with error probability:

$$\mathsf{e}_n = \tfrac{1}{2}\left[1 - (1 - 2\mathsf{e})^n\right].$$

[3] That is, for the metrics Δ and Adv_X. The tightness may not extend to other metrics, such as the mutual information between $\boldsymbol{L}(\boldsymbol{X})$ and X.

Using this, we define an equivalent leakage function:

$$\mathsf{L}'(X) = (x_1 \oplus e_1') \parallel (x_0 \oplus e_0'),$$

where $e_i' \sim \mathrm{Ber}(\mathsf{e}_n)$. Then, by applying the same structure as in (5), we get:

$$\begin{aligned}\Delta &= \mathsf{SD}(X\,;\,X \mid \boldsymbol{L(X)}) = \mathsf{SD}(X\,;\,X \mid \mathsf{L}'(X)) \\ &= \left(\tfrac{1}{2} - \mathsf{e}_n\right)\left(\tfrac{3}{2} - \mathsf{e}_n\right) \\ &= 2^{n-1}\left(\tfrac{1}{2} - \mathsf{e}\right)^n \left(1 + \tfrac{1}{2}(1-2\mathsf{e})^n\right).\end{aligned}$$

In contrast, Lemma 4 gives the bound:

$$\Delta \leq 2^{n-1}\delta^n,$$

which is looser than our exact computation:

$$\begin{aligned}\Delta &= 2^{n-1}\left(\tfrac{1}{2} - \mathsf{e}\right)^n \left(1 + \tfrac{1}{2}(1-2\mathsf{e})^n\right) \\ &<_{(\mathsf{I})} 2^{n-1}\left(\tfrac{1}{2} - \mathsf{e}\right)^n \left(1 + \tfrac{1}{2}(1-2\mathsf{e})\right)^n = 2^{n-1}\delta^{n-1},\end{aligned}$$

where step (I) uses the inequality:

$$(1 + \tfrac{1}{2}t^n) < (1 + \tfrac{1}{2}t)^n \quad \text{for } 0 < t < 1,\ n > 1. \qquad \square$$

Need for More Fine-Tuned Analysis. Our findings thus far indicate that for $q = 2^u$ with $u > 1$, the standard δ-noisy to ϵ-random probing reduction—such as in the case of $\mathsf{L}(X) = \mathsf{ZV}(X)$—and indirect metric estimates (as illustrated in Example 2) introduce a non-negligible gap. The central contribution of this paper is to close this gap via a new *decomposition-based* approach.

3.2 Decomposition into Binary Subfields

The observation that exact metrics are tractable and the reduction is tight in binary fields motivates a decomposition strategy: we reduce computations in $\mathbb{F}_{2^u}$ to binary relations, where metrics can be efficiently analyzed, and then lift the results back. We begin by outlining the foundational concepts.

Consider two u-bit integers, A and B, and define their bitwise *inner product* as:

$$\langle A, B\rangle = \bigoplus_{i=0}^{u-1} a_i b_i,$$

where a_i and b_i are the ith bits of A and B, respectively.

Let $X \in \mathbb{F}_{2^u}$ be a random variable with a masked encoding $\boldsymbol{X} = (X_1, \ldots, X_n)$. For any integer $h \in \{1, \ldots, 2^u - 1\}$—interpreted via its u-bit binary representation—we can project the equality $X = X_1 \oplus \cdots \oplus X_n$ into the binary domain as:

$$\langle X, h\rangle = \langle X_1, h\rangle \oplus \cdots \oplus \langle X_n, h\rangle.$$

We will later formalize the validity of this mapping in the context of Boolean systems. For now, to illustrate its practical utility, Lemma 6 will show that if the adversary fails to recover any of the $2^u - 1$ binary projections $\langle X, h\rangle$ from the leakage $\mathsf{L}(X)$, then they cannot infer X with meaningful advantage.

Lemma 6. *Given the leakage* $\mathsf{L}(X)$ *for* $X \in \mathbb{F}_{2^u}$*, the adversary's advantage in recovering* X *satisfies:*

$$\frac{1}{2^{u-1}} \max_h \mathsf{Adv}_{\langle X,h\rangle} \;\leq\; \mathsf{Adv}_X \;\leq\; \frac{1}{2^{u-1}} \sum_{h=1}^{2^u-1} \mathsf{Adv}_{\langle X,h\rangle} \;<\; 2 \max_h \mathsf{Adv}_{\langle X,h\rangle}.$$

Here, $\mathsf{Adv}_{\langle X,h\rangle}$ *denotes the adversary's advantage in recovering the binary inner product* $\langle X, h\rangle$*.*

Proof. Let us denote $\mu_h = \mathsf{Adv}_{\langle X,h\rangle}$. Given the set $\{\mu_1, \ldots, \mu_{2^u-1}\}$, we aim to bound Adv_X from above and below.

Let $\{p_0, p_1, \ldots, p_{2^u-1}\}$ denote the posterior distribution of X given the leakage realization l. For any h, the advantage μ_h can be written as:

$$\begin{aligned}
\mu_h &= \mathop{\mathbb{E}}_l \left[\max_{\alpha \in \{0,1\}} \Pr(\langle X, h\rangle = \alpha \mid l) \right] - \frac{1}{2} \\
&= \mathop{\mathbb{E}}_l \left[\max_{\alpha \in \{0,1\}} \sum_{i \in [0, 2^u-1], \langle i,h\rangle = \alpha} p_i \right] - \frac{1}{2} \\
&= \mathop{\mathbb{E}}_l \left[\max_{\alpha \in \{0,1\}} \left(\sum_{\langle i,h\rangle = \alpha} p_i - \frac{1}{2} \right) \right] \\
&=_{(\mathsf{I})} \frac{1}{2} \mathbb{E}_l \left[\left| \sum_{\langle i,h\rangle = 0} p_i - \sum_{\langle i,h\rangle = 1} p_i \right| \right] \\
&= \frac{1}{2} \mathbb{E}_l \left[\left| \sum_{i=0}^{2^u-1} p_i (-1)^{\langle i,h\rangle} \right| \right].
\end{aligned}$$

In step (I), we used the fact that $a + b = 1 \Rightarrow |a - b| = 2(\max\{a, b\} - \frac{1}{2})$. Let us now define the following quantity:

$$\theta_h \triangleq \sum_{i=0}^{2^u-1} p_i (-1)^{\langle i,h\rangle},$$

so that $\mathbb{E}_l[|\theta_h|] = 2\mu_h$.
Now, by the inverse Walsh-Hadamard transform, we can express p_i as:

$$p_i = \frac{1}{2^u} \sum_{h=0}^{2^u-1} \theta_h (-1)^{\langle i,h\rangle} = \frac{1}{2^u} \left(\theta_0 + \sum_{h=1}^{2^u-1} \theta_h (-1)^{\langle i,h\rangle} \right),$$

and since $\theta_0 = \sum_i p_i = 1$, we have:

$$p_i - \frac{1}{2^u} = \frac{1}{2^u} \sum_{h=1}^{2^u-1} \theta_h (-1)^{\langle i,h\rangle}. \tag{6}$$

For derivation of upper bounds in the lemma, we can write:

$$\begin{aligned}
\mathsf{Adv}_X &= \mathop{\mathbb{E}}_{l}\left[\max_i p_i - \frac{1}{2^u}\right] = \frac{1}{2^u}\mathop{\mathbb{E}}_{l}\left[\max_i[\sum_{h=1}^{2^u-1}\theta_h(-1)^{\langle i,h\rangle}]\right] \\
&\le \frac{1}{2^u}\mathop{\mathbb{E}}_{l}\left[\max_i\left|\sum_{h=1}^{2^u-1}\theta_h(-1)^{\langle i,h\rangle}\right|\right] \le \frac{1}{2^u}\mathop{\mathbb{E}}_{l}\left[\sum_{h=1}^{2^u-1}|\theta_h|\right] \\
&= \frac{1}{2^u}\sum_{h=1}^{2^u-1}2\mu_h = \frac{1}{2^{u-1}}\sum_{h=1}^{2^u-1}\mu_h < 2\max_h\ \mu_h.
\end{aligned}$$

To derive the lower bound, let $h^* = \operatorname{argmax}_h \mu_h$, and define the random index $J \in \{0,\dots,2^u-1\}$ uniformly sampled from:

$$\{J \in [0,2^u-1] \mid (-1)^{\langle J,h^*\rangle} = \operatorname{sign}(\theta_{h^*})\}.$$

We have:

$$\begin{aligned}
\mathsf{Adv}_X &= \frac{1}{2^u}\mathop{\mathbb{E}}_{l}\left[\max_i\sum_{h=1}^{2^u-1}\theta_h(-1)^{\langle i,h\rangle}\right] \\
&\ge \frac{1}{2^u}\mathop{\mathbb{E}}_{l}\left[\mathop{\mathbb{E}}_{J}\left[\sum_{h=1}^{2^u-1}\theta_h(-1)^{\langle J,h\rangle}\right]\right] \\
&= \frac{1}{2^u}\mathop{\mathbb{E}}_{l}\left[\sum_{h=1}^{2^u-1}\theta_h\mathop{\mathbb{E}}_{J}[(-1)^{\langle J,h\rangle}]\right].
\end{aligned}$$

The inner expectation simplifies because: - For $h = h^*$, we have $\mathbb{E}_J[(-1)^{\langle J,h^*\rangle}] = \operatorname{sign}(\theta_{h^*})$; - For $h \neq h^*$, since $\langle J,h\rangle$ is independent of $\langle J,h^*\rangle$, we get $\mathbb{E}_J[(-1)^{\langle J,h\rangle}] = 0$.
Therefore:

$$\mathsf{Adv}_X \ge \frac{1}{2^u}\mathop{\mathbb{E}}_{l}[\theta_{h^*}\cdot\operatorname{sign}(\theta_{h^*})] = \frac{1}{2^u}\mathop{\mathbb{E}}_{l}[|\theta_{h^*}|] = \frac{1}{2^u}\cdot 2\mu_{h^*} = \frac{\mu_{h^*}}{2^{u-1}}.$$

This completes the proof. □

3.3 Tightness of the Decomposition

We continue with the convention from the previous proof and denote $\mathsf{Adv}_{\langle X,h\rangle}$ by μ_h. The decomposition-based approach bounds the adversary's advantage by:

$$\mathsf{Adv}_X \le \frac{1}{2^{u-1}}\sum_{h\neq 0}\mu_h.$$

Our goal in this subsection is twofold: first, we show that this upper bound is tight (i.e., achievable); second, we demonstrate that for any δ-noisy leakage function, the quantity $\frac{1}{2^{u-1}}\sum_{h\neq 0}\mu_h$ remains strictly below δ.

Achievability. Consider the trivial case where the leakage function fully reveals the variable, i.e., $\mathsf{L}(X) = X$. Then, for every nonzero h, the adversary perfectly learns the binary variable $\langle X, h \rangle$, and thus $\mu_h = \frac{1}{2}$.

Substituting into the decomposition bound gives:

$$\mathsf{Adv}_X \leq \frac{1}{2^{u-1}} \cdot \left((2^u - 1) \cdot \frac{1}{2} \right) = 1 - \frac{1}{2^u}.$$

On the other hand, since the adversary recovers X exactly, we have:

$$\mathsf{Adv}_X = \Pr(\hat{X} = X) - \frac{1}{2^u} = 1 - \frac{1}{2^u}.$$

Hence, the bound is tight in this case.

Lemma 7. *Let $X \in \mathbb{F}_{2^u}$ be a uniform variable with leakage function $\mathsf{L}(X)$ such that $\mathsf{SD}(X\,;\,X \mid \mathsf{L}(X)) = \delta$. Then:*

$$\frac{1}{2^{u-1}} \sum_{h=1}^{2^u-1} \mu_h < \delta.$$

Proof. Let $\{p_0, p_1, \ldots, p_{2^u-1}\}$ denote the posterior distribution of X given a leakage instance l. Since $\langle X, h \rangle$ is a binary random variable, we can express μ_h as:

$$\mu_h = \frac{1}{2} \mathop{\mathbb{E}}_{l} \left[\left| \sum_{\langle i,h \rangle = 0} p_i - \frac{1}{2} \right| + \left| \sum_{\langle i,h \rangle = 1} p_i - \frac{1}{2} \right| \right]. \tag{7}$$

Let us define:

$$q_i \triangleq p_i - \frac{1}{2^u}, \quad \text{so that} \quad \sum_i q_i = 0.$$

Then, the definitions of δ and μ_h become:

$$\delta = \frac{1}{2} \mathop{\mathbb{E}}_{l} \left[\sum_i |q_i| \right], \quad \mu_h = \frac{1}{2} \mathop{\mathbb{E}}_{l} \left[\left| \sum_{\langle i,h \rangle = 0} q_i \right| + \left| \sum_{\langle i,h \rangle = 1} q_i \right| \right]. \tag{8}$$

While the triangle inequality immediately gives $\mu_h \leq \delta$, we aim to show that the average over all nonzero h is smaller than $\frac{1}{2}\delta$. This requires a refined argument.

Partitioning Strategy. We begin by partitioning each $q_\beta > 0$ into non-negative components:

$$q_\beta = \sum_{\alpha=0}^{2^u-1} a_{\beta,\alpha}, \quad \text{where } a_{\beta,\alpha} \geq 0.$$

These components are then redistributed to offset the negative entries, by defining:

$$q_\alpha = -\sum_\beta a_{\beta,\alpha}, \quad \text{for all } q_\alpha \leq 0.$$

	q_3	q_4	q_5	q_6	q_7
q_0	$a_{0,3}$	$a_{0,4}$	$a_{0,5}$	$a_{0,6}$	$a_{0,7}$
q_1	$a_{1,3}$	$a_{1,4}$	$a_{1,5}$	$a_{1,6}$	$a_{1,7}$
q_2	$a_{2,3}$	$a_{2,4}$	$a_{2,5}$	$a_{2,6}$	$a_{2,7}$

We illustrate this partitioning strategy with a simple example. Let $u = 3$, and suppose that q_0, q_1, q_2 are positive while q_3 through q_7 are negative. The partitioning can be visualized as the following table:

For brevity, we omit the details of how this partitioning is constructed.

Now consider the effect on absolute value terms such as $|q_\beta + q_\alpha + c|$, where $q_\beta > 0$, $q_\alpha \leq 0$, and c is the sum of the remaining terms. We can show that $|q_\beta + q_\alpha| \leq |q_\beta| + |q_\alpha| - 2a_{\beta,\alpha}$, and we derive:

$$|q_\beta + q_\alpha + c| \leq |q_\beta + q_\alpha| + |c| \leq |q_\beta| + |q_\alpha| - 2a_{\beta,\alpha} + |c|.$$

Loss Factor from Pairwise Cancellation. Applying this to all sums in μ_h, we observe that each pair $(q_\beta > 0, q_\alpha < 0)$ appears in the same half of the partition (either 0 or 1 side) for half of the nonzero h's. Hence, each such pair contributes a "loss" of at least:

$$\frac{2^u - 1}{2} \cdot 2a_{\beta,\alpha}.$$

Summing over all such pairs yields a total loss of:

$$(2^u - 1) \sum_{\beta,\alpha} a_{\beta,\alpha} = (2^u - 1) \sum_{\alpha, q_\alpha < 0} (-q_\alpha),$$

where we used $\sum_\beta a_{\beta,\alpha} = -q_\alpha$. Since $\sum_i q_i = 0$, the total positive mass equals the total negative mass:

$$\sum_{\alpha, q_\alpha < 0} (-q_\alpha) = \sum_{\beta, q_\beta > 0} q_\beta = \frac{1}{2} \sum_i |q_i|.$$

Final Bound. Combining everything:

$$\begin{aligned}\sum_{h=1}^{2^u-1} \mu_h &\leq \frac{2^u - 1}{2} \mathop{\mathbb{E}}_l \left[\sum_i |q_i| \right] - \frac{2^u - 1}{2} \cdot \frac{1}{2} \mathop{\mathbb{E}}_l \left[\sum_i |q_i| \right] \\ &= \frac{2^u - 1}{4} \mathop{\mathbb{E}}_l \left[\sum_i |q_i| \right] = \frac{2^u - 1}{2} \delta.\end{aligned}$$

Dividing both sides by 2^{u-1}, we get:

$$\frac{1}{2^{u-1}} \sum_{h=1}^{2^u-1} \mu_h \leq \frac{2^u - 1}{2^u} \delta < \delta,$$

which concludes the proof. $\square$

3.4 Application of Decomposition to Mask Encodings

Lemma 6 expresses the side-channel security of a variable $X \in \mathbb{F}_{2^u}$ in terms of the security of its binary projections $\langle X, h\rangle$. This decomposition reduces the complex task of estimating $\mathsf{Adv}_X[l \leftarrow \mathsf{L}(X)]$ to the simpler computation of the binary advantages $\mu_h = \mathsf{Adv}_{\langle X,h\rangle}[l \leftarrow \mathsf{L}(X)]$. The strength of this approach becomes especially apparent when applied to masked encodings.

Consider the setting of masked encoding, where a secret X is shared as $\boldsymbol{X} = (X_1, \ldots, X_n)$, and the adversary observes $\boldsymbol{L(X)} = (\mathsf{L}(X_1), \ldots, \mathsf{L}(X_n))$. For each nonzero $h \in \{1, \ldots, 2^u - 1\}$, the inner product satisfies the binary relation:

$$\langle X, h\rangle = \langle X_1, h\rangle \oplus \cdots \oplus \langle X_n, h\rangle.$$

That is, $\langle X, h\rangle$ is itself masked by the shares $\langle X_i, h\rangle$, making it a binary secret with a standard masked encoding.

Applying Lemma 5, the advantage of the adversary in recovering $\langle X, h\rangle$ from the masked leakage is given by:

$$\mathsf{Adv}_{\langle X,h\rangle}[\boldsymbol{l} \leftarrow \boldsymbol{L(X)}] = 2^{n-1}\left(\mathsf{Adv}_{\langle X,h\rangle}[l \leftarrow \mathsf{L}(X)]\right)^n = 2^{n-1}\left(\mu_h\right)^n,$$

where $\mu_h = \mathsf{Adv}_{\langle X_i,h\rangle}[l \leftarrow \mathsf{L}(X_i)]$ is the advantage from a single share.

Combining this with the upper bound from Lemma 6, we obtain:

$$\mathsf{Adv}_X[\boldsymbol{l} \leftarrow \boldsymbol{L(X)}] \leq \frac{1}{2^{u-1}} \sum_{h=1}^{2^u-1} 2^{n-1}\mu_h^n = \frac{1}{2^u}\sum_{h=1}^{2^u-1}(2\mu_h)^n <_{(\mathsf{I})} \Delta \leq_{(\mathsf{II})} 2^{n-1}\delta^n,$$

where:

- $\Delta = \mathsf{SD}(X\,;\, X \mid \boldsymbol{L(X)})$;
- (I) follows from Lemma 7, applied to the masked case;
- (II) follows from Lemma 4.

For clarity, we restate the result as a theorem.

Theorem 1. *Let μ_h denote the adversary's advantage in predicting $\langle X, h\rangle$ from the leakage $\mathsf{L}(X)$, for $h \in \{1, \ldots, 2^u - 1\}$. Then, in a masked encoding with share leakage $\boldsymbol{L(X)}$, the adversary's advantage satisfies:*

$$\frac{1}{2^u}\max_h(2\mu_h)^n \;\leq\; \mathsf{Adv}_X[\boldsymbol{l} \leftarrow \boldsymbol{L(X)}] \;\leq\; \frac{1}{2^u}\sum_{h=1}^{2^u-1}(2\mu_h)^n \;<\; \max_h(2\mu_h)^n.$$

Interpretation in Terms of Mutual Information Theorem 1 shows that if $2\mu_h < 1$ for all h, then the adversary's advantage Adv_X decays exponentially with the number of shares n. In this subsection, we connect the condition $\mu_h < \frac{1}{2}$ to the mutual information between the binary projection $\langle X, h\rangle$ and the leakage $\mathsf{L}(X)$. Specifically, we show that:

$$\mathsf{MI}\big(\langle X, h\rangle\,;\,\mathsf{L}(X)\big) < 1 \quad\Longrightarrow\quad \mu_h < \tfrac{1}{2}.$$

By definition of mutual information for a binary variable [9], we have:

$$\mathsf{MI}\big(\langle X,h\rangle;\, \mathsf{L}(X)\big) \;=\; 1 \;-\; \mathsf{H}\left(\tfrac{1}{2} \pm \mu_h\right), \tag{9}$$

where $\mathsf{H}(\cdot)$ denotes the binary entropy function.[4]

The mutual information reaches its maximum value of 1 only when $\mathsf{H}(\frac{1}{2} \pm \mu_h) = 0$, i.e., when $\mu_h = \frac{1}{2}$. This implies that the leakage $\mathsf{L}(X)$ fully determines $\langle X,h\rangle$. Therefore, whenever

$$\mathsf{MI}\big(\langle X,h\rangle;\, \mathsf{L}(X)\big) < 1,$$

the corresponding advantage μ_h must satisfy $\mu_h < \frac{1}{2}$, ensuring that masking offers meaningful security for the binary component $\langle X,h\rangle$.

Revisiting Previous Examples

Example 3. We revisit Example 2 to highlight the strength of the decomposition approach. For $X \in \mathbb{F}_{2^2}$ with leakage $\mathsf{L}(X) = (x_1 \oplus e_1) \parallel (x_0 \oplus e_0)$, where $\Pr(e_0 = 1) = \Pr(e_1 = 1) = \mathsf{e}$, we had:

$$\delta = \mathsf{SD}(X\,;\, X \mid \mathsf{L}(X)) = \left(\tfrac{1}{2} - \mathsf{e}\right)\left(\tfrac{3}{2} - \mathsf{e}\right).$$

Under mask encoding, we previously derived:

$$\varDelta = \mathsf{SD}(X\,;\, X \mid \boldsymbol{L(X)}) = 2^{n-1}\left(\tfrac{1}{2} - \mathsf{e}\right)^n \left(1 + \tfrac{1}{2}(1 - 2\mathsf{e})^n\right).$$

For this leakage, Example 1 (with $u = 1$) yields:

$$\mu_1 = \mu_2 = \tfrac{1}{2} - \mathsf{e},$$

corresponding to the adversary's advantage in predicting x_0 and x_1, respectively. Similarly, for μ_3, which corresponds to predicting $x_0 \oplus x_1$, we compute:

$$\mu_3 = \tfrac{1}{2} - 2\mathsf{e}(1 - \mathsf{e}).$$

By Lemma 6, the decomposition-based bound for unmasked X becomes:

$$\mathsf{Adv}_X[l \leftarrow \mathsf{L}(X)] \le \tfrac{1}{2}(\mu_1 + \mu_2 + \mu_3) = \left(\tfrac{1}{2} - \mathsf{e}\right)\left(\tfrac{3}{2} - \mathsf{e}\right) = \delta.$$

For masked encoding, Theorem 1 gives:

$$\mathsf{Adv}_X[\boldsymbol{l} \leftarrow \boldsymbol{L(X)}] \le 2^{n-2}\left(\mu_1^n + \mu_2^n + \mu_3^n\right) = \varDelta.$$

[4] The mutual information of a Binary Symmetric Channel (BSC) between A and B is

$$\mathsf{MI}(A;B) = \mathsf{H}(A) - \mathsf{H}(A \mid B) = 1 - \mathsf{H}(P_e),$$

where P_e is the probability of incorrectly estimating A given B. In our setting, $\mu_h = |P_e - \frac{1}{2}|$, so $P_e = \frac{1}{2} \pm \mu_h$.

This confirms the tightness of our bound. For instance, setting $\mathsf{e} = 0.1$, we get $\delta = 0.56$. Since $\delta > \frac{1}{2}$, Lemma 4 cannot guarantee masking security. Meanwhile, mutual information each share is $\mathsf{MI}(X_i; \mathsf{L}(X_i)) = 1.06$, which exceeds the 0.72 threshold required by Ito et al. [21] for masking to be secure. Thus, neither criterion provides a conclusive answer—while our method confirms that masking is indeed secure in this case. □

Example 4. Recall the leakage function $\mathsf{ZV}(X)$, previously introduced as:

$$\mathsf{ZV}(X) = \begin{cases} \nu_a & \text{if } X = 0, \\ \nu_b \neq \nu_a & \text{otherwise.} \end{cases}$$

For this model, (1) yields $\epsilon_{\min} = 1$, rendering the δ-noisy to ϵ-random probing reduction inapplicable for analyzing masking security.

In contrast, applying our decomposition approach, we compute each $\mu_h = \frac{1}{2^u}$ using relation (7). Then, by Theorem 1, the adversary's advantage under masking satisfies:

$$\mathsf{Adv}_X[\boldsymbol{l} \leftarrow \boldsymbol{L}(\boldsymbol{X})] \leq 2^{n-u} \sum_{h=1}^{2^u-1} \mu_h^n = (2^u - 1)\, 2^{n-u-nu},$$

which decreases exponentially with n, provided $u > 1$. Thus, even though the standard reduction fails, our approach confirms that masking remains effective in this setting. □

Application to Leakage Certification. Leakage certification laboratories evaluate cryptographic implementations on physical devices to assess their resistance against side-channel attacks (see [12, 29]). These evaluations often rely on estimating metrics such as $\mathsf{MI}(X\,;\,\mathsf{L}(X))$ and $\mathsf{SD}(X\,;\,X \mid \mathsf{L}(X))$. Estimating these quantities requires knowledge of the joint distribution $(X, \mathsf{L}(X))$, which can be derived either through parametric models (e.g., Gaussian with estimated parameters) or non-parametric methods (e.g., histogram-based) [2, 17].

Our work introduces an additional criterion for leakage assessment. Specifically, for a u-bit variable X, we propose verifying the condition:

$$\mathsf{MI}\big(\langle X, h\rangle\,;\,\mathsf{L}(X)\big) < 1 \quad \text{for all } h \in [1, 2^u - 1].$$

Masking provides meaningful side-channel protection if and only if this condition holds for every nontrivial bitwise projection $\langle X, h\rangle$.

Using Equation (9), we can express the bounds from Theorem 1 in terms of mutual information:

$$\frac{1}{2^u}\left(2\max_h \left|\tfrac{1}{2} - \mathsf{H}^{-1}(I_h)\right|\right)^n \;\leq\; \mathsf{Adv}_X[\boldsymbol{l} \leftarrow \boldsymbol{L}(\boldsymbol{X})] \;<\; \left(2\max_h \left|\tfrac{1}{2} - \mathsf{H}^{-1}\big(I_h\big)\right|\right)^n,$$

where $I_h = 1 - \mathsf{MI}\big(\langle X, h\rangle\,;\,\mathsf{L}(X)\big)$, and H^{-1} is the inverse of the binary entropy function.

3.5 Experimental Results and Determining Masking Order

We perform our experiments on a ChipWhisperer CW308 board hosting an STM32F303 UFO processor,[5] running an unprotected 8-bit software implementation of AES (included in the ChipWhisperer package). Power traces are captured using a ChipWhisperer-Lite, synchronized at a sampling rate four times the target's clock frequency.

The targeted intermediate value is the output of the first S-box in the first round:

$$X = \mathsf{S\text{-}box}(P[0] \oplus K[0]).$$

A correlation power analysis (CPA) using only 20 traces successfully recovers the secret byte, indicating a low noise level and motivating the need to assess the feasibility and necessary order of masking.

To this end, we estimate the bias parameters μ_h through profiling. Specifically, we train a deep neural network to predict X from power traces. The network processes one-dimensional input traces and has the following architecture:

- A 1D convolutional layer with 32 filters of size 3, followed by ReLU activation.
- A max-pooling layer with pool size 2.
- A second 1D convolutional layer with 64 filters of size 3, followed by ReLU activation.
- Another max-pooling layer with pool size 2.
- A flattening layer, followed by a dense layer with 128 ReLU-activated neurons.
- A final dense output layer with 256 neurons (corresponding to all possible S-box output values), followed by softmax activation.

We trained this model on 10,000 labeled traces with ground truth $X = \mathsf{S\text{-}box}(P[0] \oplus K[0])$. In the attack phase, the model's output corresponds to the posterior distribution of X given the leakage in a trace.

Using this posterior, we estimate the binary error probability P_e for computing the bitwise projection $\langle X, h\rangle = \bigoplus_{i=0}^{7} x_i h_i$ for each $h \in \{1, \dots, 255\}$. The neural network demonstrates high accuracy in estimating these projections.

Estimating P_e. For a given attack trace, let the predicted value be $\hat{X}$ and the true value be X. For each h, we compute the frequency with which $\langle \hat{X}, h\rangle = \langle X, h\rangle$ happens. The error is defined as:

$$P_e = 1 - \Pr[\langle \hat{X}, h\rangle = \langle X, h\rangle].$$

Figure 1 shows the estimated P_e for all values of h, based on 1,000 attack traces. The minimum error occurs at $h = 255$, which corresponds to Hamming weight leakage computed by $\mathsf{mod}(\mathsf{HW}(X), 2)$. The corresponding (maximum) bias is:

$$\mu_h = (1 - P_e) - \tfrac{1}{2} = 0.32.$$

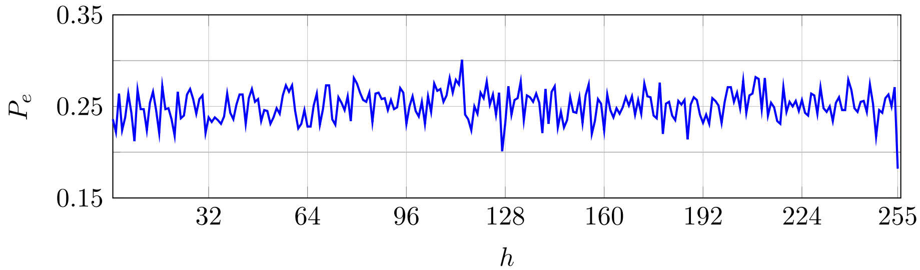


Fig. 1: Binary error probability P_e in estimating $\langle X, h \rangle$.

Substituting this value into the bound from Theorem 1, we obtain the following security bounds for masking in this setting:

$$\frac{1}{256} \cdot (0.64)^n \;\leq\; \mathsf{Adv}_X[\boldsymbol{l} \leftarrow \boldsymbol{L(X)}] \;<\; (0.64)^n.$$

These bounds illustrate that while the adversary retains a noticeable advantage for small n, increasing the masking order rapidly improves security.

4 Masking in Odd Prime Fields

Grassi et al. [18], building on earlier results by Dziembowski et al. [14], showed that masking becomes ineffective in the presence of highly informative leakage functions—such as $\mathsf{L}(X) = \mathsf{HW}(X)$—unless the masking is performed over an odd prime field. This insight has sparked further investigation into the security properties of masking in prime fields [15].

In this section, we contribute to this line of work with the following findings:

- In Subsection 4.1, we observe that the adversary's advantage Adv_X decays more rapidly in prime fields.
- In Subsection 4.2, we establish a general condition under which masking guarantees a strict reduction in Adv_X.
- In Subsection 4.3, we show that in prime fields, Adv_X decreases exponentially with n as long as $\min_h \mathsf{MI}(\langle X, h \rangle; \mathsf{L}(X)) < 1$. This condition implies that there exists at least one projection not fully leaked to the adversary—a substantially more relaxed requirement compared to the case of binary extension fields.

4.1 Adv_X Decays Faster in Prime Fields

We define a leakage class as *symmetric* if, for a uniform $X \in \mathbb{F}_q$, observing an instance l of the leakage transforms the posterior distribution $X \mid l$ into

[5] `https://github.com/newaetech/chipwhisperer`

$(p_{e_0}, p_{e_1}, \ldots, p_{e_{q-1}})$, where $\sum_{i=0}^{q-1} p_{e_i} = 1$ and p_{e_i} denotes the probability mass on the value $X + i$.[6] In this notation, p_{e_0} represents the probability of a correct estimation, and the adversary's advantage is:

$$\mathsf{Adv}_X = p_{e_0} - \frac{1}{q}.$$

Consider a masked encoding with $n = 2$, where X_1 and X_2 are the shares of X such that $X = X_1 + X_2$. Upon observing the leakages, the adversary forms estimates $\hat{X}_1$ and $\hat{X}_2$, and reconstructs X via $\hat{X}_1 + \hat{X}_2$. The reconstruction is correct if either both estimates are correct or the errors in $\hat{X}_1$ and $\hat{X}_2$ cancel out (i.e., are i and $q - i$). Hence, the updated success probability becomes:

$$p'_{e_0} = (p_{e_0})^2 + \sum_{i=1}^{q-1} p_{e_i} p_{e_{q-i}}. \tag{10}$$

Lemma 8. *For the defined symmetric leakage class, when the field order q is prime, Adv_X decays faster with increasing n.*

Proof. In a prime field, the elements i and $q - i$ are distinct for all $1 \leq i \leq \frac{q-1}{2}$, so we can rewrite Equation (10) as:

$$p'_{e_0} = (p_{e_0})^2 + 2 \sum_{i=1}^{\frac{q-1}{2}} p_{e_i} p_{e_{q-i}}.$$

In contrast, when $q = 2^u$, we have $q - i = i$, so the equation simplifies to:

$$p'_{e_0} = (p_{e_0})^2 + \sum_{i=1}^{q-1} p_{e_i}^2.$$

By applying the inequality $2ab \leq a^2 + b^2$, we obtain:

$$(p_{e_0})^2 + 2 \sum_{i=1}^{\frac{q-1}{2}} p_{e_i} p_{e_{q-i}} \leq (p_{e_0})^2 + \sum_{i=1}^{q-1} p_{e_i}^2.$$

This shows that p'_{e_0}, and hence the adversary's success probability and advantage, are lower in prime fields than in binary extension fields. $\square$

4.2 When Does Adv_X Decrease with Masking?

For a masked encoding of the secret X, we show that if there is no *hole* in the posterior distribution of the shares after observing the leakage vector, then Adv_X strictly decreases.

[6] We refer to this as a symmetric leakage class because it generalizes the binary symmetric channel.

Definitions. For later reference, we define a leakage instance l as *dummy* if it induces no change in the distribution of $X \mid l$.[7] Under dummy leakage, the peak point of the posterior distribution is $\frac{1}{q}$. A *hole* in a distribution is an element of its domain with zero probability mass. The *support* of a random variable refers to the number of domain elements with non-zero probability mass, and we denote the support size of X by $|X|$.

Problem Statement. Let X_1 and X_2 be shares of X in $\mathbb{F}_q$, and suppose the adversary observes leakage instances $l_1 \leftarrow \mathsf{L}(X_1)$ and $l_2 \leftarrow \mathsf{L}(X_2)$. The resulting posterior distributions are denoted by $\mathcal{P}^1 = (p_0^1, \ldots, p_{q-1}^1)$ and $\mathcal{P}^2 = (p_0^2, \ldots, p_{q-1}^2)$, respectively. Let $p_{i^*}^1$ and $p_{j^*}^2$ denote the peak probabilities in these distributions.

To estimate the value of X, a maximum a posteriori (MAP) adversary computes the distribution of the sum $(X_1 \mid l_1) + (X_2 \mid l_2)$, denoted by $\mathcal{P} = (p_0, \ldots, p_{q-1})$. The adversary outputs the index of the peak point of $\mathcal{P}$ as $\hat{X}$, and the corresponding advantage is denoted $\mathsf{Adv}_X[l_1, l_2 \leftarrow \mathsf{L}(X_1), \mathsf{L}(X_2)]$. Our goal is to identify conditions ensuring:

$$\mathsf{Adv}_X[l_1, l_2 \leftarrow \mathsf{L}(X_1), \mathsf{L}(X_2)] < \mathsf{Adv}_{X_i}[l_i \leftarrow \mathsf{L}(X_i)],$$

which implies that masking strictly improves the side-channel security of X.

Lemma 9. *If, for at least one non-dummy leakage instance, the posterior distributions $\mathcal{P}^1$ and $\mathcal{P}^2$ have no holes—that is, $\min \mathcal{P}^1 > 0$ and $\min \mathcal{P}^2 > 0$—then the adversary's advantage strictly decreases.*

Proof. For $0 \leq i < q$, define $\zeta_i = \frac{p_i^1}{p_{i^*}^1}$ and $\xi_i = \frac{p_i^2}{p_{j^*}^2}$. Since $p_{i^*}^1$ and $p_{j^*}^2$ are the maximum values, we have $\zeta_i \leq 1$ and $\xi_i \leq 1$. From the normalization $\sum_i p_i^1 = 1$, we obtain:

$$p_{i^*}^1 = \frac{1}{\sum_i \zeta_i}, \quad p_{j^*}^2 = \frac{1}{\sum_i \xi_i}.$$

Let k^* be the index of the peak of $\mathcal{P}$. We show that $p_{k^*} \leq \min\{p_{i^*}^1, p_{j^*}^2\}$. Without loss of generality, assume $\min = p_{i^*}^1$. Then:

$$p_{k^*} = \sum_i p_{k^*-i}^1 \, p_i^2 = p_{i^*}^1 p_{j^*}^2 \sum_i \zeta_{k^*-i} \, \xi_i \leq p_{i^*}^1,$$

because $p_{i^*}^1 > 0$ and each $\zeta_{k^*-i} \leq 1$. The inequality holds since:

$$\sum_i \zeta_{k^*-i} \, \xi_i \leq \sum_i \xi_i.$$

Equality would require:

$$\forall i, \quad \zeta_{k^*-i} = 1, \tag{11}$$

[7] We refer to it as dummy because any random variable that is independent of X has a similar effect.

which implies $p^1_i = p^1_{i^*}$ for all i, hence a uniform distribution: $p^1_{i^*} = \frac{1}{q}$. This contradicts the assumption that the leakage is non-dummy. Thus, the strict inequality $p_{k^*} < \min\{p^1_{i^*}, p^2_{j^*}\}$ must hold for at least one instance.

Therefore, taking expectations over all leakage instances, we obtain:

$$\mathbb{E}_l[p_{k^*}] < \mathbb{E}_l[p^1_{i^*}] = \mathbb{E}_l[p^2_{j^*}] \quad \Rightarrow \quad \mathsf{Adv}_X[\boldsymbol{l} \leftarrow \boldsymbol{L(X)}] < \mathsf{Adv}_X[l \leftarrow \mathsf{L}(X)]. \quad \square$$

Reaching a Hole-Free Posterior Distribution. When the leakage function is less noisy—e.g., $\mathsf{L}(X) = \mathsf{HW}(X)$—the posterior distribution of $X_i \mid \mathsf{L}(X_i)$ typically contains holes, preventing the application of Lemma 9. However, increasing the number of shares can eliminate such holes, as we now explain.

Although Lemma 9 applies directly to the case $n = 2$, the result generalizes naturally to higher n. Let $X_1, \ldots, X_{2n}$ be shares of X, with corresponding leakages $\mathsf{L}(X_1), \ldots, \mathsf{L}(X_{2n})$. The task of estimating X from its leakage vector can be decomposed into two stages: first, estimating $S_1 = X_1 + \cdots + X_n$ and $S_2 = X_{n+1} + \cdots + X_{2n}$, and then estimating $X = S_1 + S_2$.

If, beyond a certain threshold n_0, the distribution of $S_1 \mid \mathsf{L}(X_1), \ldots, \mathsf{L}(X_n)$ has no holes, then Lemma 9 implies that Adv_X will decrease with n, indicating security improvement through masking.

Because the shares and leakages are independent, the distribution of $S_1 \mid (\mathsf{L}(X_1), \ldots, \mathsf{L}(X_n))$ equals the convolution $X_1 \mid l_1 + \cdots + X_n \mid l_n$. At any leakage instance, each $X_i \mid l_i$ is a distribution, and our goal is to determine when the support of the sum reaches the full domain $\mathbb{F}_q$, which ensures a hole-free distribution.

Lemma 10 (Generalized Cauchy-Davenport Theorem). *Let $Z_1, \ldots, Z_t$ be independent random variables with supports $|Z_1|, \ldots, |Z_t|$ over a prime field $\mathbb{F}_q$. Then:*

$$|Z_1 + \cdots + Z_t| \geq \min\{|Z_1| + \cdots + |Z_t| - t + 1,\ q\}.$$

Proof. This is a direct generalization of the classical Cauchy-Davenport theorem, which states the result for $t = 2$. The case $t = 2$ was previously used in the side-channel literature by Dziembowski et al. [14]. $\square$

Using this lemma, we conclude that if each $X_i \mid l_i$ has support size greater than 1 (i.e., $|X_i \mid l_i| > 1$), then there exists some $n \geq n_0$ for which the convolution $X_1 \mid l_1 + \cdots + X_n \mid l_n$ is hole-free. Consequently, Adv_X decreases with increasing n.

4.3 Advantage in Terms of $\langle X, h\rangle$ Projections

In the case of fields with characteristic two ($q = 2^u$), we previously reduced the security of masking to the security of binary projections $\langle X, h\rangle$. In this subsection, we explore the implications of these projections for the security of masking in prime fields. To this end, we rely on the following result derived from a bound by Dziembowski et al. [14].

Lemma 11 ([14], Theorem 1). *Let $\mathbb{F}_q$ be a prime field, and suppose the leakage is δ-noisy. That is,* $\mathsf{SD}(X\,;\,X \mid \mathsf{L}(X)) = \delta < 1 - \frac{1}{q}$*. Then:*

$$\varDelta \leq \frac{1}{q} \cdot 2^{-\frac{n\sigma^4}{2}},$$

where $\varDelta = \mathsf{SD}(X\,;\,X \mid \boldsymbol{L(X)})$ *and* $\sigma = (1 - \frac{1}{q}) - \delta$.

A straightforward corollary (using Lemma 1) yields:

$$\mathsf{Adv}_X[\boldsymbol{l} \leftarrow \boldsymbol{L(X)}] \leq \frac{1}{q} \cdot 2^{-\frac{n\sigma^4}{2}}. \tag{12}$$

Now, suppose there exists h such that the leakage $\mathsf{L}(X)$ does not fully determine $\langle X, h\rangle$. That is, the bias μ_h of this bit is bounded by $0 \leq \mu_h < \frac{1}{2}$. In the borderline case where $\mu_h \to \frac{1}{2}$, we can approximate the statistical distance as follows:[8]

$$\mathsf{SD}(X\,;\,X \mid \mathsf{L}(X)) \;\lessapprox\; \frac{1}{2} + \mu_h - \frac{1}{q}.$$

Substituting into the bound (12), we obtain:

$$\mathsf{Adv}_X[\boldsymbol{l} \leftarrow \boldsymbol{L(X)}] \leq \frac{1}{q} \cdot 2^{-\frac{n(1/2-\mu_h)^4}{2}}, \tag{13}$$

which indicates that the adversary's advantage decreases exponentially in n as long as $\mu_h < \frac{1}{2}$.

We emphasize that the restriction to a single projection is made solely for illustrative purposes; the bound provided by Lemma 11 remains tighter and more general.

5 General Requirement for Secure Masking

For a finite abelian group $\mathbb{G}$, Dziembowski et al. [14] showed that there exists a leakage with noise parameter $\delta_0 = 1 - \frac{|\mathbb{H}|}{|\mathbb{G}|}$ that renders masking ineffective, where $\mathbb{H}$ denotes a largest proper (nontrivial) subgroup of $\mathbb{G}$. They further conjectured that for any leakage with $\delta < \delta_0$, masking should be effective. In this subsection we prove this conjecture for the practically relevant cyclic case $\mathbb{G} = \mathbb{Z}_N$ (with arbitrary $N \in \mathbb{N}$) and derive an explicit, exponentially decaying advantage bound.

Theorem 2. *Let X be uniform on $\mathbb{Z}_N$ and let $\mathbb{H} \leq \mathbb{Z}_N$ be a largest proper subgroup (so $[\mathbb{Z}_N : \mathbb{H}] = p$, the* smallest *prime dividing N). If, for some $\sigma > 0$, the leakage satisfies*

$$\delta \;\leq\; 1 - \frac{|\mathbb{H}|}{|\mathbb{Z}_N|} - \sigma \;=\; 1 - \frac{1}{p} - \sigma,$$

[8] This approximation assumes a worst-case scenario where the only uncertainty remaining about X after observing $\mathsf{L}(X)$ lies in the value of $\langle X, h\rangle$, which takes values in $\{0, 1\}$ with probabilities $\frac{1}{2} \pm \mu_h$.

then for every leakage outcome $\boldsymbol{l}$*,*

$$\mathsf{Adv}_X[\boldsymbol{l} \leftarrow \boldsymbol{L}(X)] \;\le\; \frac{N-1}{N}\,\rho(\sigma,p)^{\,n},$$

where

$$\rho(\sigma,p) \;\le\; \Big|(1-\sigma)+\sigma\, e^{-j\frac{2\pi}{p}}\Big| \;=\; \sqrt{1-2\sigma\Big(1-\cos\frac{2\pi}{p}\Big)+O(\sigma^2)}\,.$$

In particular, for small σ*,*

$$\rho(\sigma,p) \;\le\; 1-\Big(1-\cos\frac{2\pi}{p}\Big)\,\sigma+O(\sigma^2).$$

Proof. Fix a leakage vector $\boldsymbol{l}=(l_1,\ldots,l_n)$. For each i, let $\mathcal{P}^i=\{p_t^i\}_{t=0}^{N-1}$ be the posterior distribution of X_i given l_i, and set

$$\theta_k^i \;\triangleq\; \sum_{t=0}^{N-1} p_t^i\, e^{-j\frac{2\pi tk}{N}}, \qquad k=0,\ldots,N-1.$$

Then $\theta_0^i=1$ and $|\theta_k^i|\le 1$. Writing $\boldsymbol{P}=\mathcal{P}^1 * \cdots * \mathcal{P}^n$, convolution (denoted by "$*$" operation) becomes pointwise multiplication in the Fourier domain:

$$\Theta_k \;\triangleq\; \sum_{t=0}^{N-1} \boldsymbol{P}(t)\, e^{-j\frac{2\pi tk}{N}} \;=\; \prod_{i=1}^{n}\theta_k^i.$$

Inverting the transform, we have:

$$\boldsymbol{P}(t)=\frac{1}{N}\sum_{k=0}^{N-1}\Theta_k\, e^{j\frac{2\pi tk}{N}} \;=\; \frac{1}{N}+\frac{1}{N}\sum_{k=1}^{N-1}\Theta_k\, e^{j\frac{2\pi tk}{N}}.$$

Applying the triangle inequality gives

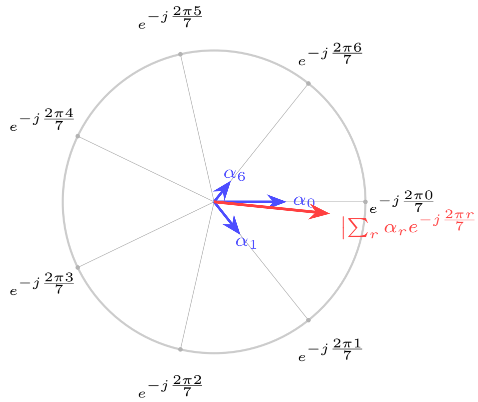
$$\mathsf{Adv}_X[\boldsymbol{l}\leftarrow\boldsymbol{L}(X)]=\max_t\Big|\boldsymbol{P}(t)-\tfrac{1}{N}\Big| \;\le\; \frac{1}{N}\sum_{k=1}^{N-1}\Big|\prod_{i=1}^{n}\theta_k^i\Big| \;\le\; \frac{N-1}{N}\Big(\max_{i,k\ne 0}|\theta_k^i|\Big)^n.$$

It remains to upper bound $\max_{i,k\neq 0}|\theta_k^i|$ in terms of δ.

Fix i and $k\neq 0$, and consider the map $t\mapsto kt \bmod N$, which partitions $\mathbb{Z}_N$ into $m=N/\gcd(N,k)$ residue classes. Pushing $\mathcal{P}^i$ forward along this map does not increase the total variation distance, so on the m-point quotient we obtain a distribution $\alpha=(\alpha_0,\ldots,\alpha_{m-1})$ with $\mathsf{SD}(\alpha,U_m)\le\delta$, and

$$|\theta_k^i|=\Big|\sum_{t=0}^{N-1}p_t^i\, e^{-j\frac{2\pi tk}{N}}\Big|=\Big|\sum_{r=0}^{m-1}\alpha_r\, e^{-j\frac{2\pi r}{m}}\Big|.$$

For instance, with $m=7$, the following diagram shows how the maximization problem looks like.



Among nontrivial quotients, the worst case (largest possible modulus under the same TV budget) occurs for the minimal size $m = p = [\mathbb{Z}_N : \mathbb{H}]$. On this p-gon, moving the available total variation σ from the "main" direction to its nearest neighbor maximizes the modulus to first order. Concretely, take

$$\alpha_0 = 1 - \sigma, \qquad \alpha_1 = \sigma, \qquad \alpha_r = 0 \ (r \geq 2).$$

A direct calculation shows that for $\sigma \leq 1/p$ its total variation from U_p is

$$\mathrm{SD}(\alpha, U_p) = \frac{1}{2}\Big(\Big|(1-\sigma) - \frac{1}{p}\Big| + \Big|\sigma - \frac{1}{p}\Big| + (p-2)\cdot\frac{1}{p}\Big) = 1 - \frac{1}{p} - \sigma \ \geq \delta,$$

so the construction is admissible under the hypothesis. Its Fourier modulus equals

$$\Big|\sum_{r=0}^{p-1} \alpha_r \, e^{-j\frac{2\pi r}{p}}\Big| = \Big|(1-\sigma) + \sigma e^{-j\frac{2\pi}{p}}\Big| = \sqrt{1 - 2\sigma\Big(1 - \cos\frac{2\pi}{p}\Big) + O(\sigma^2)}\,.$$

This yields the asserted bound on $\rho(\sigma, p)$, with tightness for $\sigma \leq 1/p$. The Taylor expansion at $\sigma = 0$ gives the linearized form $\rho(\sigma, p) \leq 1 - (1 - \cos\frac{2\pi}{p})\sigma + O(\sigma^2)$.

With larger σ, more α_i components will be involved and this will decrease the overall absolute sum, as those α_i values are in other directions. Hence, we have $\max_{i,k\neq 0} |\theta_k^i| \leq \rho(\sigma, p)$. Plugging this into the derived bound for $\mathsf{Adv}_X[\boldsymbol{l} \leftarrow \boldsymbol{L}(X)]$ completes the proof. □

5.1 Comparison with Known Bounds

Case $N = 2^u$. In this setting, the index of the largest proper subgroup is $p = [\mathbb{Z}_N : \mathbb{H}] = 2$. The corresponding quotient group therefore has only two frequencies, namely 1 and $e^{-j\pi} = -1$. Substituting $\delta = 1 - \frac{1}{2} - \sigma$ into Theorem 2, the bound simplifies to

$$\mathsf{Adv}_X[\boldsymbol{l} \leftarrow \boldsymbol{L}(X)] \ \leq \ \frac{2^u - 1}{2^u}\,(2\delta)^n.$$

This expression coincides with the bounds established earlier in Lemma 5 and Lemma 4, confirming the tightness of our general result in the binary case.

Case $N = q$ with q prime. When N is prime, we have $p = q$. In this case, the dedicated bound from Lemma 11 is tighter than the general bound derived in Theorem 2. The reason is that, in our general derivation, an additional multiplicative factor of $(N-1)$ appears in the advantage bound. While negligible for small q, this factor makes our bound looser compared to the prime-field bound, particularly when q is large.

6 Security of Linear Gadgets

Our analysis so far has focused on standalone secrets and their masked encodings. We now extend this to more complex structures, specifically linear *gadgets*. Our goal is to demonstrate the applicability of the proposed decomposition technique for evaluating security in such settings.

Gates and Gadgets. A gadget is a *family* of circuits (one per order n) that securely implements a masked version of a gate. Let $\mathsf{G}\colon (\mathbb{F}_{2^u})^t \to \mathbb{F}_{2^u}$ be a gate with fan-in t. For example, XOR and AND have $t = 2$. The corresponding gadget, denoted $\mathbb{S}\mathsf{G}\colon (\mathbb{F}_{2^u}^n)^t \to \mathbb{F}_{2^u}^n$, takes masked inputs and returns masked outputs.

A *refresh gadget* has fan-in and fan-out 1. It re-randomizes the encoding of a value $\boldsymbol{X}$ while preserving its underlying secret, i.e., $\bigoplus_i X_i = \bigoplus_i X_i'$. A well-known example is the $\mathbb{S}\mathsf{R}\text{-}\mathsf{SNI}$ gadget [1], shown in Algorithm 1.

Algorithm 1 $\mathbb{S}\mathsf{R}\text{-}\mathsf{SNI}$

Input $\boldsymbol{X} = (X_1, \ldots, X_n)$
Output $\boldsymbol{X}' = (X_1', \ldots, X_n')$

for $i = 1$ **to** n **do**
  for $j = i + 1$ **to** n **do**
    $r \xleftarrow{\$} \mathbb{F}_{2^u}$
    $X_i = X_i \oplus r$
    $X_j = X_j \oplus r$
return $\boldsymbol{X}' = \boldsymbol{X}$

6.1 Problem Statement

Let $\boldsymbol{\Sigma}_n = \{V_1, \ldots, V_{\mathsf{T}(n)}\}$ be the set of intermediate variables in an $\mathbb{F}_2$-linear gadget processing secret $X \in \mathbb{F}_{2^u}$. Besides the shares of X, $\boldsymbol{\Sigma}_n$ may include random values whose leakage could help an adversary estimate X.

The linearity assumption implies the existence of a matrix $\boldsymbol{P}_n \in \mathbb{F}_2^{\mathsf{P}(n)\times(\mathsf{T}(n)+1)}$ such that:

$$\boldsymbol{P}_n \cdot [X, V_1, \ldots, V_{\mathsf{T}(n)}]^\dagger = \mathbf{0}. \tag{14}$$

This system describes all parity constraints among the variables, and any additional relations are linear combinations of these rows. The adversary is assumed to know $\boldsymbol{P}_n$ (or any equivalent form).

Side-channel measurements yield:

$$\boldsymbol{L}_n = [\mathsf{L}(V_1), \ldots, \mathsf{L}(V_{\mathsf{T}(n)})],$$

and the adversary's goal is to estimate X from $\boldsymbol{L}_n$ and $\boldsymbol{P}_n$. We denote their advantage as $\mathsf{Adv}_X[\boldsymbol{l}_n \leftarrow \boldsymbol{L}_n]$.

MAP Adversary and Exact Advantage. Let $\mathcal{S}_n$ be the set of all solutions to (14). Each $\boldsymbol{S} \in \mathcal{S}_n$ is a vector of length $\mathsf{T}(n)+1$, with $\boldsymbol{S}(0)$ denoting the value of X. The MAP adversary outputs:

$$\hat{X} = \underset{\alpha \in \mathbb{F}_{2^u}}{\operatorname{argmax}} \sum_{\substack{\boldsymbol{S} \in \mathcal{S}_n \\ \boldsymbol{S}(0)=\alpha}} \Pr(\boldsymbol{S} \mid \boldsymbol{l}_n).$$

The corresponding advantage is:

$$\mathsf{Adv}_X[\boldsymbol{l}_n \leftarrow \boldsymbol{L}_n] = \mathbb{E}_{\boldsymbol{l}_n}\left[\Pr(\hat{X} = X \mid \boldsymbol{l}_n)\right] - \frac{1}{2^u}.$$

A Non-Tight Upper Bound. To obtain a computable upper bound, we apply the reduction from δ-noisy leakage to ϵ-random probing. We replace each leakage $\mathsf{L}(V_i)$ with its erasure version $\phi^\epsilon(V_i)$, where $\epsilon \geq \epsilon_{\min}$ and $\epsilon_{\min}$ is derived from L (cf. (1)). This yields:

$$\mathsf{Adv}_X[\boldsymbol{l}_n \leftarrow \boldsymbol{L}_n] \leq \mathsf{Adv}_X[l_1, \ldots, l_{\mathsf{T}(n)} \leftarrow \phi^\epsilon(V_1), \ldots, \phi^\epsilon(V_{\mathsf{T}(n)})].$$

We denote this bound by $\mathsf{Adv}(q, n, \epsilon)$, where $q = 2^u$. Since smaller ϵ yields less information, we have:

$$\mathsf{Adv}_X[\boldsymbol{l}_n \leftarrow \boldsymbol{L}_n] \leq \mathsf{Adv}(q, n, \epsilon_{\min}) \leq \mathsf{Adv}(q, n, \epsilon). \tag{15}$$

Jahandideh et al. [22] estimated $\mathsf{Adv}(q, n, \epsilon)$ for various gadgets. For the $\mathbb{S}\mathsf{R}$-SNI gadget with $n < 30$ and $\epsilon < 0.15$, they showed:

$$\mathsf{Adv}(q, n, \epsilon) \leq \frac{q-1}{q} \cdot \epsilon^{0.6n}. \tag{16}$$

However, for some leakage functions—e.g., $\mathsf{L}(X) = \mathsf{ZV}(X)$—the reduction gives $\epsilon_{\min} = 1$, leading to a trivial bound:

$$\mathsf{Adv}_X[\boldsymbol{l}_n \leftarrow \boldsymbol{L}_n] \leq 1 - \frac{1}{q}.$$

In the next subsection, we show how our decomposition approach avoids this issue and yields tighter bounds in $\mathbb{F}_{2^u}$ settings.

6.2 Decomposition into Binary Systems

In an $\mathbb{F}_2$-linear system such as

$$\boldsymbol{P}_n \cdot [X, V_1, \ldots, V_{\mathsf{T}(n)}]^\dagger = \mathbf{0},$$

the projections $\langle X, h\rangle$ and $\langle V_i, h\rangle$ also satisfy the same linear structure. More precisely:

Lemma 12. *Let $h \in \{1, \ldots, 2^u - 1\}$. Then the system $\boldsymbol{P}_n \cdot [X, V_1, \ldots, V_{\mathsf{T}(n)}]^\dagger = \mathbf{0}$ implies:*

$$\boldsymbol{P}_n \cdot [\langle X, h\rangle, \langle V_1, h\rangle, \ldots, \langle V_{\mathsf{T}(n)}, h\rangle]^\dagger = \mathbf{0}.$$

Proof. Using the linearity of the binary inner product:

$$\langle V_1 \oplus V_2, h\rangle = \langle V_1, h\rangle \oplus \langle V_2, h\rangle, \quad \langle bV, h\rangle = b \cdot \langle V, h\rangle$$

for $b \in \mathbb{F}_2$, the result follows by applying these identities to each row of $\boldsymbol{P}_n$. $\square$

A Tighter Upper Bound for $\mathsf{Adv}_{\boldsymbol{X}}[\boldsymbol{l_n} \leftarrow \boldsymbol{L_n}]$. The adversary also obtains side-channel leakage from each intermediate variable. From Lemma 6, we derive the following bound for the secret X:

$$\mathsf{Adv}_X[\boldsymbol{l}_n \leftarrow \boldsymbol{L}_n] \leq \frac{1}{2^{u-1}} \sum_{h=1}^{2^u-1} \mathsf{Adv}_{\langle X,h\rangle}[\boldsymbol{l}_n \leftarrow \boldsymbol{L}_n], \tag{17}$$

where each term $\mathsf{Adv}_{\langle X,h\rangle}[\boldsymbol{l}_n \leftarrow \boldsymbol{L}_n]$ corresponds to the adversary's advantage in estimating a single binary projection $\langle X, h\rangle$ under δ-noisy leakage.

Applying the reduction from δ-noisy leakage to ϵ-random probing, we obtain:

$$\mathsf{Adv}_{\langle X,h\rangle}[\boldsymbol{l}_n \leftarrow \boldsymbol{L}_n] \leq \mathsf{Adv}(2, n, \epsilon^h_{\min}),$$

where $\epsilon^h_{\min}$ depends on both the projection index h and the leakage function $\mathsf{L}(V)$. Importantly, while the systems for X and $\langle X, h\rangle$ are structurally the same, the field size is reduced to $\mathbb{F}_2$.

For binary variables, we previously established that:

$$\epsilon^h_{\min} = 2\mu_h, \tag{18}$$

where μ_h denotes the bias in estimating $\langle V, h\rangle$. One way to compute μ_h is via mutual information. As discussed in Section 3.4, it can be expressed as:

$$\mu_h = \left| \frac{1}{2} - \mathsf{H}^{-1}\left[1 - \mathsf{MI}\big(\langle V, h\rangle\,;\, \mathsf{L}(V)\big)\right] \right|,$$

where H^{-1} is the inverse of the binary entropy function.

Putting all of this together, we obtain:

$$\mathsf{Adv}_X[\boldsymbol{l}_n \leftarrow \boldsymbol{L}_n] \leq \frac{1}{2^{u-1}} \sum_{h=1}^{2^u-1} \mathsf{Adv}(2, n, \epsilon^h_{\min}) \leq 2 \cdot \max_h \mathsf{Adv}(2, n, \epsilon^h_{\min}). \tag{19}$$

This shows that the decomposition approach enables a tighter reduction from the δ-noisy to the ϵ-random probing model for $\mathbb{F}_{2^u}$-valued linear circuits.

Example 5. To illustrate the usefulness of the upper bound in (19), consider the leakage function $\mathsf{ZV}(V)$ and the $\mathbb{S}$R-SNI gadget. As discussed in Example 4, for $\mathsf{ZV}(V)$, we have $\mu_h = \frac{1}{2^u}$, implying:

$$\epsilon^h_{\min} = 2\mu_h = \frac{1}{2^{u-1}}.$$

Applying the bound from (16), we obtain:

$$\mathsf{Adv}_X[\boldsymbol{l}_n \leftarrow \boldsymbol{L}_n] \leq \mathsf{Adv}(2, n, 2\mu_h) \leq \frac{1}{2}\left(\frac{1}{2^{u-2}}\right)^{0.6n},$$

which holds as long as $4\mu_h < 0.15$, i.e., $u \geq 5$.

Without the decomposition approach introduced in this work, the minimum erasure rate would be $\epsilon_{\min} = 1$, yielding a trivial upper bound of $\mathsf{Adv}_X[\boldsymbol{l}_n \leftarrow \boldsymbol{L}_n] \leq 1$. □

7 Conclusion

This work establishes a unified and tight characterization of the noise thresholds required for masking to be effective. By introducing a decomposition approach that reduces leakage over extended fields to binary projections, we derived tight bounds on the adversary's success rate and demonstrated that these bounds yield an optimal reduction from the noisy leakage model to the random probing model, even in low-noise regimes where classical reductions collapse.

A central outcome of our analysis is a proof of the conjecture of Dziembowski et al. (TCC 2016), showing that for an additive group $\mathbb{G}$ with largest proper subgroup $\mathbb{H}$, any δ-noisy leakage with $\delta < 1 - \frac{|\mathbb{H}|}{|\mathbb{G}|}$ ensures that masking strictly improves security. Our results generalize previous thresholds for binary fields and provides, for the first time, necessary and sufficient conditions for masking in arbitrary additive groups.

Beyond its theoretical contributions, our framework has direct practical relevance. We demonstrated how the decomposition strategy refines leakage evaluation by identifying which bitwise projections carry residual information, enabling more accurate certification of implementations. Our experimental evaluation confirmed that the proposed criteria not only capture realistic low-noise leakage scenarios but also allow determining the required masking order to achieve a given security target.

Overall, our results close a longstanding gap in the theoretical foundations of masking, strengthen the link between noisy leakage and probing models, and provide concrete tools for the design and certification of side-channel resistant implementations. Future work includes extension of the decomposition approach to non-linear gadgets and exploring its applicability on other masking schemes such as *inner-product masking*.

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